



**YEAR 11 Mathematics Ext 1  
Preliminary Course  
Assessment Task 1  
March 2009**

1. There are 3 sections.
2. Answer each question on your own paper showing all necessary working
3. Start each section on a new page
4. Calculators may be used

| Topic                       | Mark |
|-----------------------------|------|
| 1. Section 1 (Algebra)      | /40  |
| 2. Section 2 (Polynomials)  | /16  |
| 3. Section 3 (Inequalities) | /16  |

**TOTAL                      /72**

**SECTION 1 ALGEBRA**  
(Total for this section is 40 marks)

Marks

1. Simplify  $2^x + 2^x$  (1)
2. Simplify  $\frac{2^{n+1} - 2^{n-1}}{2^n}$  (2)
3. Simplify  $\left(\frac{a+1}{a^3+1}\right)^3$  (2)
4. Simplify  $\frac{4^m \times 27^{m-n}}{6^{2m}}$  (3)
5. Solve  $2^{3m-1} = \frac{1}{4^m}$  (2)
6. Factorise  $12(x^2 - y^2) - 7xy$  (3)
7. Factorise  $n^4 + 4m^4$  (HINT: add and subtract  $4m^2n^2$ ) (3)
8. Solve  $x^{\frac{3}{2}} = \frac{1}{8}$  (2)
9. Factorise fully  $x^5 - 9x^3 - 8x^2 + 72$  and hence solve  

$$x^5 - 9x^3 - 8x^2 + 72 = 0$$
 (4)
10. Express  $\frac{1 - x^{-1}}{x^{-1} - x^{-2}}$  in simplest form (3)
11. Simplify  $\frac{x^3 - (x-y)^3}{x^2 - (x-y)^2}$  (3)
12. If  $a + \frac{x}{b} = b + \frac{x}{a}$  (and  $a \neq b$ ), find  $x$  in simplest form (3)
13. Solve simultaneously  $y = (3-x)^3$   
 $y = (3-x)(3+x)$  (4)
14. Write in simplest form  $(x^{\frac{1}{2}} + x^{\frac{-1}{2}})^2$  (2)
15. If  $a^x = b$  and  $a^{2x} = b^3$ , find  $b$  and  $x$  (3)

**SECTION 2 Polynomials**  
(Total for this section is 16 marks)

Marks

1. From the following list of expressions, state which ones are polynomials (3)
- a)  $x^2 + 7x - 3$
  - b)  $3 - \frac{1}{x}$
  - c)  $x\sqrt{x+4x}$
  - d) 5
  - e)  $9x - x^3 + x^5$
  - f)  $2^x + 1$
2. For the polynomial  $P(x) = 2x^3 + x^2 - x + 3$  state the (3)
- a) degree
  - b) leading coefficient
  - c) constant term
3. If  $P(x) = x^2 + 2x + 3$ , find (4)
- a)  $P(1)$
  - b)  $P(1) - P(-1)$
  - c)  $P(0)$
  - d) State why  $P(x) = 0$  has no real solutions
4. If  $P(x) = x^3 + 2x^2 + 3x + 2$  and  $Q(x) = x + 1$ , find (6)
- a)  $P(x) + Q(x)$
  - b)  $P(x) - Q(x)$
  - c)  $P(x) \times Q(x)$
  - d)  $P(x) \div Q(x)$

### SECTION 3 Inequalities with the unknown in the denominator

(Total for this section is 16 marks)

Marks

1. Solve  $\frac{4}{x-1} > 1$

(3)

2. Solve  $\frac{4}{(x-1)^2} \geq 1$

(4)

3. Solve  $\frac{2x}{x+1} < x$

(4)

4. Solve  $-1 \leq \frac{2x}{x+1} \leq 1$

(5)

**(END OF EXAMINATION)**

## Part A (Algebra) Solutions

1.  $2 \cdot 2^x = 2^{x+1}$

2.  $\frac{2^n(2-2^{-1})}{2^n} = 2 - \frac{1}{2}$   
 $= 1\frac{1}{2}$

3.  $\frac{(a+1)^3}{(a+1)(a^2-a+1)} = \frac{(a+1)^2}{a^2-a+1}$

4.  $\frac{4^m \times 27^m \times 27^{-n}}{36^m} = 3^m \times 3^{-3n}$   
 $= 3^{m-3n}$

5.  $2^{3m-1} = \frac{1}{2^{2m}}$   
 $2^{3m-1} = 2^{-2m}$   
 $3m-1 = -2m$   
 $5m = 1$   
 $m = \frac{1}{5}$

6.  $12x^2 - 7xy - 12y^2$   
 $\begin{array}{ccc} 4x & & 3y \\ & \searrow & \nearrow \\ & & -4y \\ 3x & \nearrow & \searrow \end{array}$   $(4x+3y)(3x-4y)$

$$7. \quad n^4 + 4m^2n^2 + 4m^4 - 4m^2n^2$$

$$\begin{array}{ccc} n^2 & & 2m^2 \\ & \diagdown & \diagup \\ & \times & \\ & \diagup & \diagdown \\ n^2 & & 2m^2 \end{array}$$

$$= (n^2 + 2m^2)^2 - (2mn)^2$$

$$= [n^2 + 2m^2 - 2mn][n^2 + 2m^2 + 2mn]$$

$$8. \quad x^{\frac{3}{2}} = \frac{1}{8}$$

$$x = \left(\frac{1}{8}\right)^{\frac{2}{3}}$$

$$x = \frac{1}{4}$$

$$\begin{aligned}
 9. \quad & x^5 - 9x^3 - 8x^2 + 72 \\
 &= x^3(x^2 - 9) - 8(x^2 - 9) \\
 &= (x^2 - 9)(x^3 - 8) \\
 &= (x - 3)(x + 3)(x - 2)(x^2 + 2x + 4)
 \end{aligned}$$

$$\begin{aligned}
 \text{Solve } (x^2 - 9)(x^3 - 8) &= 0 \\
 x &= \pm 3 \text{ and } x = 2
 \end{aligned}$$

$$10. \quad \frac{1 - \frac{1}{x}}{\frac{1}{x} - \frac{1}{x^2}} = \frac{\frac{x-1}{x}}{\frac{x-1}{x^2}}$$

$$= \frac{x-1}{x} \times \frac{x^2}{x-1}$$

$$= x$$

$$11. \quad \frac{[x - (x-y)][x^2 + x(x-y) + (x-y)^2]}{[x - (x-y)][x + (x-y)]}$$

$$= \frac{x^2 + x^2 - xy + x^2 - 2xy + y^2}{2x - y}$$

$$= \frac{3x^2 - 3xy + y^2}{2x - y}$$

$$12. \quad \frac{x}{b} - \frac{x}{a} = b - a$$

$$x\left(\frac{1}{b} - \frac{1}{a}\right) = b - a$$

$$x\left(\frac{a-b}{ab}\right) = b - a$$

$$x = (b-a) \div \frac{(a-b)}{ab}$$

$$= \frac{b-a}{1} \times \frac{ab}{-(b-a)}$$

$$= -ab \quad \checkmark$$

Since  $a \neq b$

$$13. \quad (3-x)^3 = (3-x)(3+x)$$

$$(3-x)^3 - (3-x)(3+x) = 0$$

$$(3-x)[(3-x)^2 - (3+x)] = 0$$

$$(3-x)[9 - 6x + x^2 - 3 - x] = 0$$

$$(3-x)[x^2 - 7x + 6] = 0$$

$$x = 3 \quad \text{and} \quad (x-6)(x-1) = 0$$

$$x = 6 \quad \text{and} \quad 1$$

$$\therefore x = 1, 3, 6 \quad y = 8, 0, -27 \quad /$$

$$14. \quad x + 2x^0 + x^{-1} = x + \frac{1}{x} + 2 \quad /$$

$$\text{OR} \quad \frac{x^2 + 2x + 1}{x} = \frac{(x+1)^2}{x} \quad \checkmark$$

## Part B (polynomials)

1.  $x^2 + 7x - 3$ ,  $5$ ,  $9x - x^3 + x^5$

2. Degree = 3  
leading coefficient = 2  
constant term = 3

3.  $P(x) = x^2 + 2x + 3$

(a)  $P(1) = (1)^2 + 2(1) + 3$   
 $= 6$

(b)  $P(1) - P(-1) = 6 - [(-1)^2 + 2(-1) + 3]$   
 $= 6 - 2$   
 $= 4$

(c)  $P(0) = 3$

(d)  $x^2 + 2x + 3 = 0$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times 3}}{2}$$

$$x = \frac{-2 \pm \sqrt{-8}}{2}$$

no real solutions to  $\sqrt{-8}$

$\therefore$  no real solutions to

$$P(x) = 0$$

$$15. \quad a^x = b$$

$$a^{2x} = b^2$$

$$a^{2x} = b^3$$

$$\therefore b^3 = b^2$$

$$b^3 - b^2 = 0$$

$$b^2(b-1) = 0$$

$$b = 0 \text{ and } b = 1$$

$$\text{When } b = 0$$

$$a^x = 0$$

no solution

$$\text{When } b = 1$$

$$a^x = 1$$

$$x = 0$$

$$\therefore b = 1, x = 0 \text{ only.}$$

✓

$$4. \quad (a) \quad x^3 + 2x^2 + 3x + 2 + x + 1 \\ = x^3 + 2x^2 + 4x + 3$$

$$(b) \quad x^3 + 2x^2 + 3x + 2 - x - 1 \\ = x^3 + 2x^2 + 2x + 1$$

$$(c) \quad x(x^3 + 2x^2 + 3x + 2) + 1(x^3 + 2x^2 + 3x + 2) \\ = x^4 + 2x^3 + 3x^2 + 2x + x^3 + 2x^2 + 3x + 2 \\ = x^4 + 3x^3 + 5x^2 + 5x + 2$$

$$(d) \quad \begin{array}{r} x^2 + x + 2 \\ x+1 \overline{) x^3 + 2x^2 + 3x + 2} \\ \underline{x^3 + x^2} \phantom{+ 2} \downarrow \\ x^2 + 3x \phantom{+ 2} \phantom{\downarrow} \\ \underline{x^2 + x} \phantom{+ 2} \phantom{\downarrow} \\ 2x + 2 \phantom{\downarrow} \\ \underline{2x + 2} \\ 0 \end{array}$$

$$\therefore P(x) \div Q(x) = x^2 + x + 2$$

## Part C

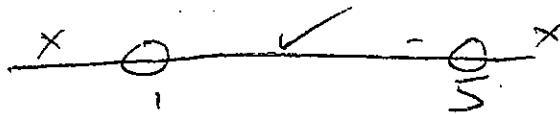
1.  $\frac{4}{x-1} > 1$  CP at  $x=1$ ,  $x \neq 1$

Make equation to find other CP's

$$4 = x-1$$

$$5 = x$$

CP at  $x=5$ ,  $x \neq 5$



$\therefore 1 < x < 5$

2.  $\frac{4}{(x-1)^2} \geq 1$  CP at  $x=1$ ,  $x \neq 1$

Make equation to find other CP's

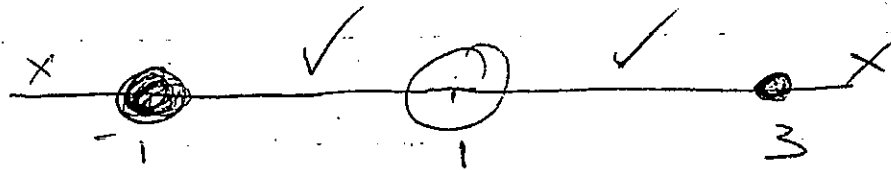
$$\frac{4}{(x-1)^2} = 1$$

$$4 = x^2 - 2x + 1$$

$$0 = x^2 - 2x - 3$$

$$0 = (x-3)(x+1)$$

CP's at  $x=3$  and  $-1$



$-1 \leq x < 1$  and  $1 < x \leq 3$

OR

$-1 \leq x \leq 3$  but  $x \neq 1$

3.  $\frac{2x}{x+1} < x$  CP at  $x = -1, x \neq -1$

Make equation to find other CP's

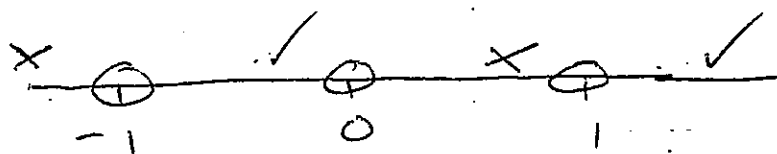
$$\frac{2x}{x+1} = x$$

$$2x = x^2 + x$$

$$0 = x^2 - x$$

$$0 = x(x-1)$$

CP at  $x = 0, x \neq 0$   
 $x = 1, x \neq 1$



$-1 < x < 0$  and  $x > 1$

4.

$$-1 \leq \frac{2x}{x+1} \leq 1$$

CP at  $x = -1, x \neq -1$

Consider two equations to find other CP's

$$-x - 1 = 2x$$

and

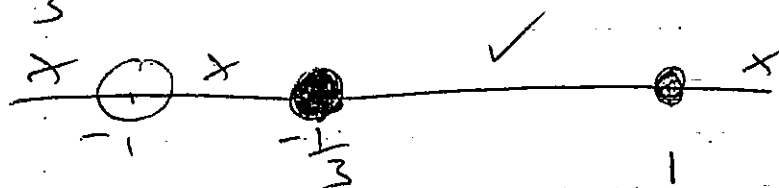
$$2x = x + 1$$

$$-1 = 3x$$

$$-\frac{1}{3} = x$$

and

$$x = 1$$



$$-\frac{1}{3} \leq x \leq 1$$